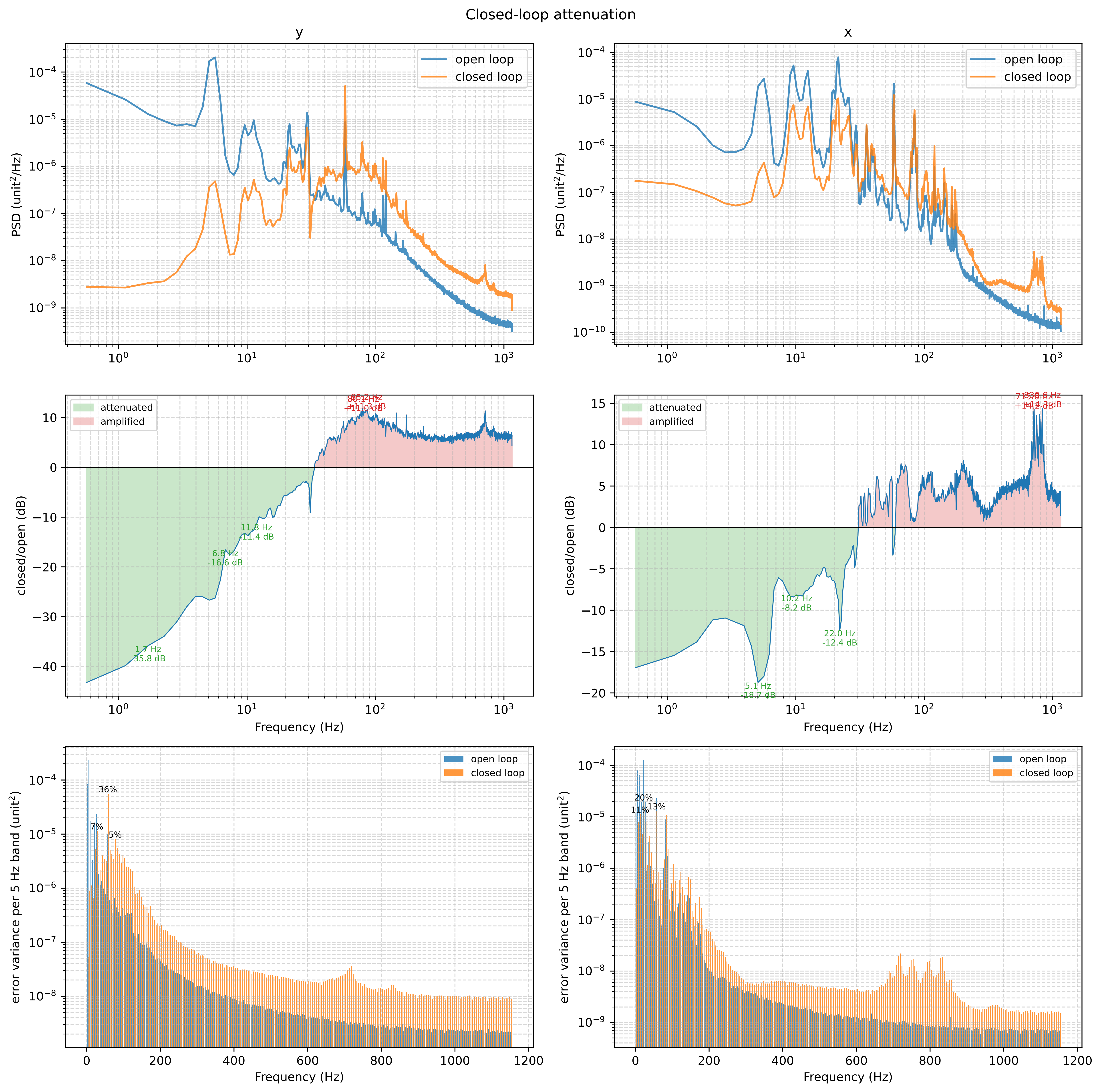
Closed-loop attenuation
y
x
open loop
closed loop
PSD (unit²/Hz)
attenuated
amplified
closed/open (dB)
Frequency (Hz)
1.7 Hz
-35.8 dB
6.8 Hz
-16.6 dB
11.8 Hz
-11.4 dB
10.2 Hz
-8.2 dB
22.0 Hz
-12.4 dB
5.1 Hz
-18.7 dB
error variance per 5 Hz band (unit²)
36%
7%
5%
20%
11%
13%

Run configuration and per-band RMS

```
Started: 2026-07-13 16:41:19
pixel_scale: 31.5 px per output unit
y: mean +0.467 -> +0.000 px;  RMS about mean 0.738 -> 0.389 px   (open -> closed)
x: mean -0.894 -> -0.001 px;  RMS about mean 0.940 -> 0.313 px   (open -> closed)

Config used:
100, 500, 100, 500, 4096: camera ASI290MM 64x64, ROI start (920,488), ~2310 Hz loop
pid: integrator-only (p=py=d=0), i=600 iy=1300, units=minmax
lines x=[22.08,57.3] gain=[60,20] phase=[56,3]
linesy=[30.8] gainy=[60] phasey=[60], line_delay=0.002
kalman: order=60 horizon=5 mu=0.03 gain=1.0 gainy=0 ramp=10
piplate: fine ch0(+2)/ch1(-2), coarse parked, baud 460800, start_delay: 100.0, open_time: 500.0, settle_time: 100.0, closed_time: 500.0
```

RMS contributed by each 10 Hz band (px)

band (Hz)	y open	y closed	x open	x closed
0-10	0.5599	0.0308	0.3016	0.0910
10-20	0.1426	0.0419	0.2907	0.1250
20-30	0.1984	0.1282	0.3796	0.1641
30-40	0.0545	0.0573	0.0640	0.0730
40-50	0.0477	0.0864	0.0397	0.0541
50-60	0.1029	0.2406	0.1512	0.1153
60-70	0.0330	0.0957	0.0192	0.0378
70-80	0.0318	0.1064	0.0328	0.0428
80-90	0.0282	0.0989	0.1021	0.1141
90-100	0.0271	0.0836	0.0154	0.0273
100-110	0.0255	0.0775	0.0224	0.0420
110-120	0.0260	0.0684	0.0158	0.0275
120-130	0.0222	0.0555	0.0227	0.0357
130-140	0.0158	0.0396	0.0148	0.0229
140-150	0.0153	0.0396	0.0238	0.0358
150-160	0.0132	0.0304	0.0099	0.0161
160-170	0.0131	0.0283	0.0103	0.0174
170-180	0.0120	0.0277	0.0117	0.0208
180-190	0.0102	0.0214	0.0062	0.0118
190-200	0.0099	0.0205	0.0051	0.0111
200-210	0.0093	0.0193	0.0044	0.0099
210-220	0.0086	0.0181	0.0040	0.0083
220-230	0.0079	0.0165	0.0037	0.0071
230-240	0.0077	0.0161	0.0039	0.0069
240-250	0.0073	0.0144	0.0037	0.0058
250-260	0.0071	0.0140	0.0034	0.0051
260-270	0.0065	0.0128	0.0031	0.0045
270-280	0.0063	0.0123	0.0031	0.0043
280-290	0.0061	0.0121	0.0031	0.0039
290-300	0.0057	0.0114	0.0028	0.0035
total 0-300	0.6294	0.3841	0.6023	0.3109